Lecture 19

Type A tensor products

Objectives. *A tensor-product multiplicity is a coefficient in a product of characters. In type A the characters are Schur polynomials. Their products are computed by adding boxes to Young diagrams.*

By the end of this lecture you will be able to:

- Extract tensor-product multiplicities by highest-weight subtraction.
- Prove Pieri's rule from the bialternant formula.
- Apply the Littlewood–Richardson rule and translate its partitions into $\mathfrak{sl}_n$ highest weights.
- Decompose the square of the adjoint module of $\mathfrak{sl}_3$ into symmetric and alternating parts.

Throughout, $\mathfrak{g}$ is a complex semisimple Lie algebra with the data fixed in Lecture 9. All modules are finite-dimensional. For type A, assume $n \geq 2$. Write $\mu \leq \lambda$ when $\lambda - \mu$ is a sum of simple roots with nonnegative integer coefficients.

19.1 Multiplicities and highest-weight subtraction

By Lecture 14, every module V has a decomposition

$$V \cong \bigoplus_{\nu \in P_+} \operatorname{Hom}_{\mathfrak{g}}(L(\nu), V) \otimes L(\nu)$$

with finitely many nonzero terms. The *multiplicity* of $L(\nu)$ in V is

$$m_\nu(V) = \dim \operatorname{Hom}_{\mathfrak{g}}(L(\nu), V).$$

For $\lambda, \mu, \nu \in P_+$, the *tensor-product multiplicities* are

$$N_{\lambda\mu}^{\nu} = m_\nu\big(L(\lambda) \otimes L(\mu)\big).$$

Characters are additive and multiplicative (Lecture 15). Hence

$$\operatorname{ch} L(\lambda)\ \operatorname{ch} L(\mu) = \sum_{\nu \in P_+} N_{\lambda\mu}^{\nu} \operatorname{ch} L(\nu).$$

Definition 19.1 (Singular vector). The singular subspace of weight ν is

$$V_\nu^{\mathfrak{n}_+} = \{v \in V_\nu : xv = 0 \text{ for all } x \in \mathfrak{n}_+\}.$$

A nonzero vector in this subspace is called *singular*, as in Lecture 13.

Lemma 19.2 (Multiplicities count singular vectors · *proved*). *Let $\nu \in P$. If $\nu \in P_+$, then* $\dim V_\nu^{\mathfrak{n}_+} = m_\nu(V)$. *If $\nu \notin P_+$, then $V_\nu^{\mathfrak{n}_+} = 0$.*

Proof. First let $\lambda \in P_+$ and take a nonzero singular vector $v \in L(\lambda)_\nu$. PBW (Lecture 11) gives

$$U(\mathfrak{g}) = U(\mathfrak{n}_-)\,U(\mathfrak{h})\,U(\mathfrak{n}_+).$$

Here $\mathfrak{n}_+$ kills v and $\mathfrak{h}$ scales it, so $U(\mathfrak{g})v = U(\mathfrak{n}_-)v$. All weights of this space are $\le \nu$. It is a nonzero submodule, hence all of $L(\lambda)$. It contains the weight λ, so $\lambda \le \nu$. Every weight of $L(\lambda)$ is $\le \lambda$ (Lecture 12), so $\nu \le \lambda$. Therefore $\nu = \lambda$ and $v \in L(\lambda)_\lambda = \mathbb{C}v_\lambda$.

Now put $M_\lambda = \operatorname{Hom}_{\mathfrak{g}}(L(\lambda), V)$. On $M_\lambda \otimes L(\lambda)$ the algebra acts on the second factor only. Weight spaces and common kernels of $\mathfrak{n}_+$ are therefore computed in that factor:

$$V_\nu^{\mathfrak{n}_+} \cong \bigoplus_{\lambda \in P_+} M_\lambda \otimes L(\lambda)_\nu^{\mathfrak{n}_+}.$$

By the first paragraph, the only possible nonzero term is $M_\nu \otimes \mathbb{C}v_\nu$. It occurs only when $\nu \in P_+$, and its dimension is $m_\nu(V)$. ∎

Proposition 19.3 (Highest-weight subtraction · *proved*). *Let $V \neq 0$.*

(i) If μ is a maximal weight of V for $\le$, then $\mu \in P_+$ and $m_\mu(V) = \dim V_\mu$.

(ii) The character $\operatorname{ch} V$ determines every multiplicity $m_\nu(V)$.

Proof. (i) Let $v \in V_\mu$ and $x \in \mathfrak{g}_\alpha$ with $\alpha \in R_+$. Then $xv \in V_{\mu+\alpha}$, and this space is zero by maximality. Thus $V_\mu = V_\mu^{\mathfrak{n}_+} \neq 0$. Lemma 19.2 gives $\mu \in P_+$ and $m_\mu(V) = \dim V_\mu$.

(ii) Use induction on $\dim V$. Choose a maximal weight μ. By the maximal-weight assertion, $m = \dim V_\mu$ is read off from $\operatorname{ch} V$. Write $V \cong L(\mu)^{\oplus m} \oplus V'$, where V' is the sum of the other isotypic components. Then

$$\operatorname{ch} V' = \operatorname{ch} V - m \operatorname{ch} L(\mu),$$

and $\operatorname{ch} L(\mu)$ is given by the Weyl character formula (Lecture 16). Moreover $m_\nu(V') = m_\nu(V)$ for $\nu \neq \mu$, and $\dim V' < \dim V$. If $V' = 0$, then $m_\nu(V) = 0$ for all $\nu \neq \mu$. Otherwise the induction hypothesis applies to V'. ∎

Characters are Weyl invariant, so it suffices to record dominant weight multiplicities. A weight maximal among the dominant weights of V is maximal among all weights. Otherwise a maximal weight lies strictly above it, and that weight is dominant by the maximal-weight assertion of Proposition 19.3. Choose such a μ and record $m_\mu(V) = \dim V_\mu$. Subtract $\dim V_\mu$ times the dominant multiplicities of $L(\mu)$ from Lecture 17, and repeat with the remainder.

19.2 Schur polynomials and tensor multiplicities

We use the dictionary of Lecture 18. A diagonal matrix $\operatorname{diag}(t_1, \dots, t_n)$ acts on a vector of $\mathfrak{gl}_n$-weight $\gamma \in \mathbb{Z}^n$ by $\sum_i \gamma_i t_i$. Put $x_i = e^{\varepsilon_i}$. A polynomial $\mathfrak{gl}_n$-module, as defined in Lecture 18, is a finite direct sum of Schur modules. The character of a polynomial $\mathfrak{gl}_n$-module V is $\sum_\gamma \dim V_\gamma\, x^\gamma$. For a partition λ with $\ell(\lambda) \le n$, the *bialternant formula* reads

$$\operatorname{ch} \mathbb{S}_\lambda(\mathbb{C}^n) = s_\lambda = \frac{a_{\lambda+\delta}}{a_\delta},$$

where $\delta = (n-1, \dots, 1, 0)$ and the *alternant* a_γ is

$$a_\gamma = \det\left(x_i^{\gamma_j}\right)_{i,j=1}^{n} = \sum_{w \in S_n} \operatorname{sgn}(w) \prod_{j=1}^{n} x_{w(j)}^{\gamma_j}.$$

The weights of $\mathbb{S}_\lambda(\mathbb{C}^n)$ lie in $\mathbb{Z}^n_{\geq 0}$ and have coordinate sum $|\lambda|$. The identity matrix acts by $|\lambda|$. Restricted to $\mathfrak{sl}_n$, the module $\mathbb{S}_\lambda(\mathbb{C}^n)$ is the irreducible $L(\bar{\lambda})$, where

$$\bar{\lambda} = \sum_{i=1}^{n-1} (\lambda_i - \lambda_{i+1})\, \omega_i.$$

Lemma 19.4 (Independence of Schur polynomials · *proved*). *The polynomials $s_\nu(x_1, \dots, x_n)$ with $\ell(\nu) \leq n$ are linearly independent.*

Proof. The exponent vectors of the monomials of a_γ are permutations of γ. For a partition ν, the vector $\nu + \delta$ is strictly decreasing. It is the only strictly decreasing permutation of itself, and its monomial has coefficient 1, from $w = \mathrm{id}$. Hence the coefficient of $x^{\nu'+\delta}$ in $a_{\nu+\delta}$ is 1 if $\nu' = \nu$ and 0 otherwise. The alternants $a_{\nu+\delta}$ are therefore independent. Division by the Vandermonde determinant $a_\delta \neq 0$ preserves independence. ∎

Proposition 19.5 (Products of Schur modules · *proved*). *Let λ, μ be partitions with at most n parts. There are unique integers $c^\nu_{\lambda\mu} \geq 0$, zero unless $|\nu| = |\lambda| + |\mu|$, with*

$$s_\lambda s_\mu = \sum_{\ell(\nu) \leq n} c^\nu_{\lambda\mu} s_\nu \quad \text{in } \mathbb{Z}[x_1, \dots, x_n]. \tag{19.1}$$

As $\mathfrak{gl}_n$-modules,

$$\mathbb{S}_\lambda(\mathbb{C}^n) \otimes \mathbb{S}_\mu(\mathbb{C}^n) \cong \bigoplus_{\ell(\nu) \leq n} \mathbb{S}_\nu(\mathbb{C}^n)^{\oplus c^\nu_{\lambda\mu}}.$$

Proof. Let $T = \mathbb{S}_\lambda(\mathbb{C}^n) \otimes \mathbb{S}_\mu(\mathbb{C}^n)$ and $d = |\lambda| + |\mu|$. The identity matrix acts on T by d, since the tensor action adds the scalars. By Weyl's theorem (Lecture 5), T is a direct sum of irreducible $\mathfrak{sl}_n$-submodules U. Since $\mathfrak{gl}_n = \mathfrak{sl}_n \oplus \mathbb{C}I$ and I acts by a scalar, each U is a $\mathfrak{gl}_n$-submodule.

Let v be a highest weight vector of U. It is a weight vector for the diagonal matrices of $\mathfrak{gl}_n$; call its weight ν. The weights of T are sums of weights of the two factors, so $\nu \in \mathbb{Z}^n_{\geq 0}$ and $|\nu| = d$. Dominance for $\mathfrak{sl}_n$ gives $\nu_i - \nu_{i+1} \geq 0$. Thus ν is a partition with $\ell(\nu) \leq n$. Both U and $\mathbb{S}_\nu(\mathbb{C}^n)$ are irreducible $\mathfrak{sl}_n$-modules of highest weight $\bar{\nu}$, so they are isomorphic (Lecture 13). The isomorphism commutes with I, which acts by d on both, so it is an isomorphism of $\mathfrak{gl}_n$-modules.

Hence $T \cong \bigoplus \mathbb{S}_\nu(\mathbb{C}^n)^{\oplus m_\nu}$ with $m_\nu \geq 0$ and $|\nu| = d$. Taking characters gives $s_\lambda s_\mu = \sum m_\nu s_\nu$. Uniqueness follows from Lemma 19.4. ∎

The integers $c^\nu_{\lambda\mu}$ are the *Littlewood–Richardson coefficients*. Since $s_\lambda s_\mu = s_\mu s_\lambda$, they satisfy $c^\nu_{\lambda\mu} = c^\nu_{\mu\lambda}$.

Corollary 19.6 (Multiplicities for $\mathfrak{sl}_n$ · *proved*). *With λ, μ as above,*

$$L(\bar{\lambda}) \otimes L(\bar{\mu}) \cong \bigoplus_{\substack{\ell(\nu) \leq n \\ |\nu| = |\lambda| + |\mu|}} L(\bar{\nu})^{\oplus c^\nu_{\lambda\mu}},$$

and distinct ν in this sum give distinct $\bar{\nu}$. For each ν in this sum, $N^{\bar{\nu}}_{\bar{\lambda}\bar{\mu}} = c^\nu_{\lambda\mu}$.

Proof. Restrict Proposition 19.5 to $\mathfrak{sl}_n$. If $\bar{\nu} = \bar{\nu}'$, then $\nu - \nu' = c(1, \dots, 1)$ for an integer c. Equal sizes give $nc = 0$, so $c = 0$ and $\nu = \nu'$. ∎

Determinant shifts. Even when $\lambda_n = \mu_n = 0$, a product may contain a ν with $\nu_n > 0$. By Lecture 18, as $\mathfrak{gl}_n$-modules,

$$\mathbb{S}_\nu(\mathbb{C}^n) \cong \det^{\nu_n} \otimes \mathbb{S}_{\nu - \nu_n(1^n)}(\mathbb{C}^n).$$

The determinant module is trivial on $\mathfrak{sl}_n$. Hence removing full columns gives the $\mathfrak{sl}_n$ highest weight and leaves the multiplicity unchanged.

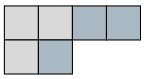
19.3 Pieri's rule

For an integer $k \geq 0$, let h_k be the sum of all monomials of degree k in $x_1, \dots, x_n$. The monomials of degree k in a basis of $\mathbb{C}^n$ form a weight basis of $\operatorname{Sym}^k(\mathbb{C}^n)$. Hence $\operatorname{ch} \operatorname{Sym}^k(\mathbb{C}^n) = h_k$.

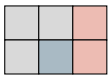
Definition 19.7 (Horizontal strip)**.** Let $\lambda \subseteq \nu$ be partitions with at most n parts. The skew diagram ν/λ is a *horizontal strip* if

$$\nu_1 \geq \lambda_1 \geq \nu_2 \geq \lambda_2 \geq \cdots \geq \nu_n \geq \lambda_n.$$

The interlacing condition says that no two boxes of ν/λ share a column. Indeed, if $\nu_i > \lambda_{i-1}$, then column ν_i has boxes of ν/λ in rows $i-1$ and i. Conversely, boxes of ν/λ in rows $j < i$ and one column c give $\lambda_{i-1} \leq \lambda_j < c \leq \nu_i$. Figure 19.1 compares an accepted shape with a rejected one, for $\lambda = (2, 1)$ and three new boxes.



$\nu = (4, 2)$
$4 \geq 2 \geq 2 \geq 1 \geq 0 \geq 0$



$\nu = (3, 3)$
$\nu_2 = 3 > \lambda_1 = 2$

Figure 19.1: Three boxes added to the grey diagram $\lambda = (2, 1)$. For $\nu = (4, 2)$ the skew diagram ν/λ is a horizontal strip, and its interlacing chain holds. The shape $\nu = (3, 3)$ is rejected, because its two red boxes lie in one column.

Theorem 19.8 (Pieri's rule · *proved*)**.** *Let λ be a partition with $\ell(\lambda) \leq n$ and let $k \geq 0$. Then*

$$s_\lambda h_k = \sum_\nu s_\nu,$$

summed over partitions ν with $\ell(\nu) \leq n$ such that ν/λ is a horizontal strip with k boxes.

Proof. Put $\beta = \lambda + \delta$, so $\beta_1 > \beta_2 > \cdots > \beta_n \geq 0$. Let A be the set of $\alpha \in \mathbb{Z}^n_{\geq 0}$ with $|\alpha| = k$.

Step 1: multiplication. For $w \in S_n$ and a polynomial f, let wf be obtained by substituting $x_{w(j)}$ for x_j. Then $a_\beta = \sum_w \operatorname{sgn}(w)\, w(x^\beta)$. Since h_k is symmetric and $h_k = \sum_{\alpha \in A} x^\alpha$,

$$\begin{aligned} h_k\, a_\beta &= \sum_{w \in S_n} \operatorname{sgn}(w)\, w\big(h_k x^\beta\big) \\ &= \sum_{\alpha \in A} \sum_{w \in S_n} \operatorname{sgn}(w)\, w\big(x^{\beta+\alpha}\big) \\ &= \sum_{\alpha \in A} a_{\beta+\alpha}. \end{aligned}$$

Step 2: cancellation. Call $\alpha \in A$ *bad* if

$$\beta_i + \alpha_i \geq \beta_{i-1} \quad \text{for some } i \in \{2, \dots, n\}.$$

For bad α, let i be the largest such index and put $\gamma = \beta + \alpha$. Let γ^* be γ with its entries in positions $i-1$ and i exchanged, and put $\alpha^* = \gamma^* - \beta$. The changed entries are

$$\alpha^*_{i-1} = \gamma_i - \beta_{i-1} \geq 0, \qquad \alpha^*_i = \gamma_{i-1} - \beta_i \geq \beta_{i-1} - \beta_i > 0.$$

So $\alpha^* \in A$. It is bad at i, since $\beta_i + \alpha^*_i = \gamma_{i-1} \geq \beta_{i-1}$. For $j > i$ the condition at j involves only γ_j and β_{j-1}, which are unchanged. Thus i is also the largest bad index of α^*, and $(\alpha^*)^* = \alpha$. Exchanging two exponents exchanges two columns of the determinant:

$$a_{\beta+\alpha^*} = a_{\gamma^*} = -a_\gamma = -a_{\beta+\alpha}.$$

Let S be the sum of $a_{\beta+\alpha}$ over bad α. Reindexing by the bijection $\alpha \mapsto \alpha^*$ gives $S = -S$, so $S = 0$.

Step 3: the surviving terms. A good α satisfies $\beta_i + \alpha_i < \beta_{i-1}$ for all $i \geq 2$. With $\nu = \lambda + \alpha$ this reads

$$\nu_i + n - i < \lambda_{i-1} + n - i + 1, \qquad \text{that is,} \qquad \nu_i \leq \lambda_{i-1}.$$

Together with $\nu_i \geq \lambda_i$, this is the horizontal-strip condition. It makes ν a partition with $|\nu| = |\lambda| + k$, and $\beta + \alpha = \nu + \delta$. Conversely, each such ν comes from the good vector $\alpha = \nu - \lambda$. Therefore

$$h_k \, a_{\lambda+\delta} = \sum_\nu a_{\nu+\delta}.$$

Dividing by a_δ gives the theorem. ∎

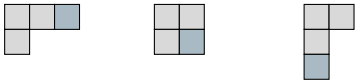
For $\lambda = \varnothing$ the only horizontal strip is $\nu = (k)$, so $h_k = s_{(k)}$. Since $\mathrm{Sym}^k(\mathbb{C}^n) = \mathbb{S}_{(k)}(\mathbb{C}^n)$ (Lecture 18), Proposition 19.5 gives the module form.

Corollary 19.9 (Pieri's rule for modules · *proved*)**.** *For $\ell(\lambda) \leq n$,*

$$\mathbb{S}_\lambda(\mathbb{C}^n) \otimes \mathrm{Sym}^k(\mathbb{C}^n) \cong \bigoplus_\nu \mathbb{S}_\nu(\mathbb{C}^n),$$

summed over the partitions ν of Theorem 19.8.

For $k = 1$ no column can repeat, so every way of adding one box to a partition gives a horizontal strip. Figure 19.2 shows the three additions to $(2, 1)$.



$(3,1)$ $\qquad$ $(2,2)$ $\qquad$ $(2,1,1)$

Figure 19.2: The one-box additions to the grey diagram $(2, 1)$, with the added box in colour and the resulting partition below.

Example 19.10 (The product $8 \otimes 3$ for $\mathfrak{sl}_3$)**.** Write $L(a, b) = L(a\omega_1 + b\omega_2)$ for $\mathfrak{sl}_3$. The adjoint module is $8 = L(1, 1) = \mathbb{S}_{(2,1)}(\mathbb{C}^3)$, and $3 = L(1, 0) = \mathbb{C}^3$. Pieri's rule gives

$$s_{(2,1)} h_1 = s_{(3,1)} + s_{(2,2)} + s_{(2,1,1)}.$$

The last partition has a full column. Removing it gives (1). Hence

$$8 \otimes 3 \cong L(2,1) \oplus L(0,2) \oplus L(1,0).$$

Lecture 17 gives

$$24 = 15 + 6 + 3.$$

A natural place to stop. The lecture has now extracted multiplicities from characters, proved Pieri's rule and decomposed $8 \otimes 3$. The next part states the Littlewood–Richardson rule for a general partition μ and uses it to decompose $8 \otimes 8$.

19.4 The Littlewood–Richardson rule

Pieri's rule multiplies by a single row. A general μ requires tableaux. Draw a partition ν as a Young diagram with ν_i boxes in row i, rows numbered downwards.

Definition 19.11 (Littlewood–Richardson tableau)**.** Let $\lambda \subseteq \nu$ and μ be partitions. A *Littlewood–Richardson tableau* of shape ν/λ and content μ is a filling of the boxes of ν not in λ by positive integers such that:

(a) entries weakly increase from left to right along each row;

(b) entries strictly increase from top to bottom down each column;

(c) exactly μ_j entries equal j, for each j;

(d) the *reverse reading word*, read from right to left along each row, from the top row to the bottom row, is a *lattice word*: every initial segment contains at least as many entries j as entries $j+1$, for every $j \geq 1$.

Theorem 19.12 (Littlewood–Richardson rule · *cited*)**.** Fulton–Harris, Appendix A, Section A.1. *Let λ, μ, ν be partitions with at most n parts, and define $c^{\nu}_{\lambda\mu}$ by (19.1). Then $c^{\nu}_{\lambda\mu}$ is the number of Littlewood–Richardson tableaux of shape ν/λ and content μ. It is 0 unless $\lambda \subseteq \nu$. It is the same for every $n \geq \max\bigl(\ell(\lambda), \ell(\mu), \ell(\nu)\bigr)$.*

Put $\lambda^{(0)} = \lambda$. For each $j \geq 1$, the boxes of λ together with the boxes with entries $\leq j$ form a partition shape $\lambda^{(j)}$. The boxes with entry j form the horizontal strip $\lambda^{(j)}/\lambda^{(j-1)}$. The rule is applied as follows.

(1) Add μ_1 boxes labelled 1 to λ as a horizontal strip.

(2) Add μ_2 boxes labelled 2 to the new shape as a horizontal strip, then μ_3 boxes labelled 3, and so on.

(3) Keep the fillings whose reverse reading word is a lattice word.

For $\mu = (k)$ every entry is 1. The lattice condition then holds automatically, and the rule reduces to Pieri's rule.

Remark 19.13 (Outer and inner products · *sketched*)**.** The coefficient $c^{\nu}_{\lambda\mu}$ also counts V^{ν} in $\operatorname{Ind}_{S_{|\lambda|}\times S_{|\mu|}}^{S_{|\lambda|+|\mu|}}(V^{\lambda} \boxtimes V^{\mu})$, the outer product of symmetric-group modules. This follows from Schur–Weyl duality and Frobenius reciprocity. The solution of Exercise 19.5 gives the argument and compares the outer product with the inner tensor product.

19.5 The square of the adjoint module of $\mathfrak{sl}_3$

We decompose $8 \otimes 8$ for $\mathfrak{sl}_3$. The modules that occur are

$$1 = L(0,0), \quad 8 = L(1,1), \quad 10 = L(3,0), \quad 10^* = L(0,3), \quad 27 = L(2,2).$$

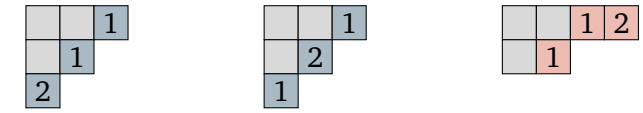
Here $10 = \mathrm{Sym}^3(\mathbb{C}^3)$.

Step 1: Littlewood–Richardson tableaux

Take $\lambda = \mu = (2,1)$ and $n \geq 4$. Content $(2,1)$ means two entries 1 and one entry 2. Adding two entries 1 to $(2,1)$ as a horizontal strip gives the shapes

$$(4,1), \qquad (3,2), \qquad (3,1,1), \qquad (2,2,1).$$

Then one entry 2 is added. The reading word must begin with 1. Figure 19.3 draws the two tableaux of shape $(3,2,1)$ and one rejected filling of shape $(4,2)$.



$\nu = (3,2,1)$ 112 $\quad$ $\nu = (3,2,1)$ 121 $\quad$ $\nu = (4,2)$ 211

Figure 19.3: Fillings of ν/λ with content $(2,1)$, for the grey diagram $\lambda = (2,1)$, with the reverse reading word below each diagram. The two fillings of $(3,2,1)$ have lattice words, and the word 211 of the red filling of $(4,2)$ is not a lattice word.

In the table, $*$ marks a box of λ and slashes separate rows. All eight surviving tableaux are listed.

ν	filling	word	$\mathfrak{sl}_3$-module
$(4,2)$	$**11 / *2$	112	27
$(4,1,1)$	$**11 / * / 2$	112	10
$(3,3)$	$**1 / *12$	121	10^*
$(3,2,1)$	$**1 / *1 / 2$	112	8
$(3,2,1)$	$**1 / *2 / 1$	121	8
$(2,2,2)$	$** / *1 / 12$	121	1
$(3,1,1,1)$	$**1 / * / 1 / 2$	112	absent
$(2,2,1,1)$	$** / *1 / 1 / 2$	112	absent

The coefficient $c^{\nu}_{(2,1),(2,1)}$ is the number of table rows with shape ν. It is 2 for $\nu = (3,2,1)$ and 1 for each of the other six shapes.

Step 2: translation to $\mathfrak{sl}_3$

Now take $n = 3$. The coefficients do not depend on n, and the two partitions with four parts drop out. Corollary 19.6 gives

$$8 \otimes 8 \cong 27 \oplus 10 \oplus 10^* \oplus 8 \oplus 8 \oplus 1, \qquad 64 = 27 + 10 + 10 + 8 + 8 + 1.$$

The partitions $(4,1,1)$, $(3,2,1)$ and $(2,2,2)$ have full columns. After the determinant shift they become (3), $(2,1)$ and $\varnothing$. As $\mathfrak{gl}_3$-modules these summands are $\det \otimes \mathrm{Sym}^3$, $\det \otimes \mathbb{S}_{(2,1)}$ and $\det^2$.

Step 3: check by highest-weight subtraction

The weights of 8 are the six roots, each with multiplicity one, and 0 with multiplicity two. A weight of $8 \otimes 8$ is a sum of two weights of 8, and its multiplicity counts ordered pairs. For example, $3\omega_1 = 2\alpha_1 + \alpha_2$, and the only pairs are

$$(\alpha_1,\ \alpha_1+\alpha_2), \qquad (\alpha_1+\alpha_2,\ \alpha_1).$$

For the highest root $\theta = \alpha_1 + \alpha_2 = \omega_1 + \omega_2$ and for 0,

$$\begin{aligned} \dim(8\otimes 8)_\theta &= \underbrace{2+2}_{(\theta,0),\,(0,\theta)} + \underbrace{2}_{(\alpha_1,\alpha_2),\,(\alpha_2,\alpha_1)} = 6, \\ \dim(8\otimes 8)_0 &= \underbrace{6}_{(\alpha,-\alpha),\ \alpha\in R} + \underbrace{2\cdot 2}_{(0,0)} = 10. \end{aligned}$$

List dominant multiplicities in the order $\big((2,2),(3,0),(0,3),(1,1),(0,0)\big)$. Kostant's formula (Lecture 17) gives

$$27 : (1,1,1,2,3), \quad 10 : (0,1,0,1,1), \quad 10^* : (0,0,1,1,1), \quad 8 : (0,0,0,1,2).$$

The subtraction algorithm runs as follows:

$$\begin{aligned} 8\otimes 8 &: (1,2,2,6,10) \\ \text{subtract } 27 &: (0,1,1,4,7) \\ \text{subtract } 10 &: (0,0,1,3,6) \\ \text{subtract } 10^* &: (0,0,0,2,5) \\ \text{subtract } 2\cdot 8 &: (0,0,0,0,1) \\ \text{subtract } 1 &: (0,0,0,0,0). \end{aligned}$$

This confirms Step 2.

Step 4: symmetric and alternating parts

The flip $V\otimes V \to V\otimes V$, $u\otimes v \mapsto v\otimes u$, is equivariant (Lecture 14). Its eigenspaces give $V\otimes V = \operatorname{Sym}^2 V \oplus \Lambda^2 V$. Choose a weight basis $v_1,\dots,v_d$ of V with weights $\mu_1,\dots,\mu_d$. The vectors v_iv_j with $i\le j$ and $v_i\wedge v_j$ with $i<j$ are weight bases of $\operatorname{Sym}^2 V$ and $\Lambda^2 V$. Both v_iv_j and $v_i\wedge v_j$ have weight $\mu_i+\mu_j$. Write $\psi^2\big(\sum_\mu m_\mu e^\mu\big) = \sum_\mu m_\mu e^{2\mu}$, so that $\psi^2 \operatorname{ch} V = \sum_i e^{2\mu_i}$. Splitting the pairs (i,j) into $i<j$, $i=j$ and $i>j$ gives

$$(\operatorname{ch} V)^2 = \sum_{i,j=1}^{d} e^{\mu_i+\mu_j} = 2\sum_{i<j} e^{\mu_i+\mu_j} + \psi^2 \operatorname{ch} V.$$

Hence

$$\begin{aligned} \operatorname{ch}\operatorname{Sym}^2 V &= \frac{1}{2}\big((\operatorname{ch} V)^2 + \psi^2 \operatorname{ch} V\big), \\ \operatorname{ch}\Lambda^2 V &= \frac{1}{2}\big((\operatorname{ch} V)^2 - \psi^2 \operatorname{ch} V\big). \end{aligned}$$

For $V = 8$, the only dominant weights of $\psi^2 \operatorname{ch} 8$ are $2\theta = (2,2)$, once, and 0, twice. In the order above,

$$\psi^2 \operatorname{ch} 8 : (1,0,0,0,2), \qquad \operatorname{Sym}^2 8 : (1,1,1,3,6), \qquad \Lambda^2 8 : (0,1,1,3,4).$$

In $\Lambda^2 8$ the maximal weights are $(3,0)$ and $(0,3)$. Subtracting 10 and 10^* leaves $(0,0,0,1,2)$, which is 8. In $\mathrm{Sym}^2 8$, subtracting 27 leaves $(0,0,0,1,3)$. Subtracting 8 then leaves the trivial module. Hence

$$\mathrm{Sym}^2 8 \cong 27 \oplus 8 \oplus 1, \qquad \Lambda^2 8 \cong 10 \oplus 10^* \oplus 8,$$

with dimensions $36 = 27 + 8 + 1$ and $28 = 10 + 10 + 8$.

Step 5: the two adjoint summands

The weight space $(8 \otimes 8)_\theta$ has dimension 6. Four of these dimensions lie in 27 (two) and in 10 and 10^* (one each). By Lemma 19.2, the singular subspace of weight θ has dimension

$$m_\theta(8 \otimes 8) = \dim \mathrm{Hom}_{\mathfrak{sl}_3}(8, 8 \otimes 8) = 2.$$

The bracket map $b\colon 8 \otimes 8 \to 8$, $b(X \otimes Y) = [X,Y]$, is equivariant by the Jacobi identity. It vanishes on $\mathrm{Sym}^2 8$, since the bracket is antisymmetric. It is nonzero:

$$b(E_{12} \otimes E_{21}) = \mathrm{diag}(1,-1,0).$$

By Schur's lemma, b is zero on $10 \oplus 10^*$. Hence b is an isomorphism on the alternating copy of 8. The symmetric copy corresponds to the symmetric cubic form of Exercise 19.6. The flip therefore splits the multiplicity space into two lines.

Self-checks

(1) Decompose $3 \otimes 3$ for $\mathfrak{sl}_3$ by Pieri's rule. Which summand is symmetric?
(2) Why is $c^{(3,2,1)}_{(2,1),(2,1)} = 2$ not the multiplicity of the corresponding weight in $8 \otimes 8$?
(3) Why does the trivial module occur exactly once in $8 \otimes 8$, and in which part?

Answers. (1) $s_{(1)}h_1 = s_{(2)} + s_{(1,1)}$, so $3 \otimes 3 \cong 6 \oplus 3^*$. Since $\mathrm{Sym}^2 \mathbb{C}^3 = \mathbb{S}_{(2)}(\mathbb{C}^3)$, the symmetric part is 6. (2) It is the dimension of the singular subspace of weight θ. The whole weight space $(8 \otimes 8)_\theta$ has dimension 6. (3) By Lecture 14, $\mathrm{Hom}_{\mathfrak{sl}_3}(\mathbb{C}, 8 \otimes 8) \cong \mathrm{Hom}_{\mathfrak{sl}_3}(8^*, 8)$. Also $8^* \cong 8$, so Schur's lemma gives dimension one. It lies in $\mathrm{Sym}^2 8$, because the invariant tensor corresponds to the invariant form $\mathrm{tr}(XY)$ under $8 \cong 8^*$, and this form is symmetric.

Summary

Tensor multiplicities are dimensions of singular subspaces. Highest-weight subtraction extracts them from characters.

In type A the multiplicities are Littlewood–Richardson coefficients, read after determinant shifts. Pieri's rule adds horizontal strips. For $\mathfrak{sl}_3$, $8 \otimes 8$ contains the adjoint module once in each symmetry type.

Reading

Fulton–Harris, §6.1, §15.3 and Appendix A.1, for Pieri and Littlewood–Richardson. Goodman–Wallach, §9.3. Sheet 19 gives further products and an invariant cubic.